HAPPIEST HEALTH DIGITAL TWIN CHALLENGE 2026

GlucoTwin India

A personalised digital twin that warns **70 minutes** before a glucose spike, and lets a doctor test diet and activity changes on the patient's virtual replica **before prescribing them**.

Team Vicimus · Vishwakarma Institute of Technology
Chitrangad Ram Sapate · Soham Joshi

THE PROBLEM

India manages diabetes by looking in the rear-view mirror

101 M

Indians living with diabetes, plus **136 M** with prediabetes (ICMR-INDIAB, 2023)

1 in 3

or fewer reach glycaemic targets. Most complications are driven by uncontrolled excursions

~26,000

CGM readings per patient between two 3-monthly OPD visits (before counting the smartwatch streams), that no doctor can read

- **HbA1c is an average.** It hides the post-meal spikes that carbohydrate-heavy Indian diets (rice, wheat) cause every day.
- **Care is reactive.** The doctor sees what already happened, never what is about to happen.
- **Advice is generic.** "Eat less rice" ignores that two patients respond to the same meal in completely different ways.

The challenge: fuse the static record (EHR) with the dynamic stream (wearables) into a model that **predicts adverse events before they happen** and that a doctor can actually use.

OUR SOLUTION

A virtual patient the doctor can query, not just a risk score

Early warning

"Spike likely in 90 min"

Forecasts glucose at 30 / 60 / 120 min with calibrated uncertainty, and flags excursions above 180 mg/dL early enough to act: a walk or a smaller portion.

What-if

"What if she eats jowar roti instead?"

The doctor simulates meals, portions and post-meal walks on **this patient's own calibrated physiology** and sees the curve before prescribing.

Explained

"Why is the risk high?"

Every alert lists its patient-specific drivers (meal, sleep debt, HRV, trend, physiology) and a pre-consultation summary is drafted automatically.

Who uses it: endocrinologists and GPs in OPD and tele-consults, diabetes educators for diet counselling. Next: patients and ASHA workers via vernacular alerts.

WHY THIS IS A DIGITAL TWIN

Five properties that separate a twin from a predictor

1 Personalised

Six physiological parameters fitted to each patient, starting from a prior built from **their EHR**.

2 Synchronised

A state observer pulls the twin toward every new CGM reading. The twin is always "now".

3 Predictive (hybrid)

ML learns what physiology leaves out (sleep debt, stress, dawn effect) from the wearable + EHR streams.

4 Simulatable

Mechanistic core answers counterfactuals that a pure ML model cannot.

5 Actionable

Triage, explanations and summaries in the doctor's workflow.

> Static EHR stream × dynamic wearable stream → a physiological replica, updated every 5 minutes.

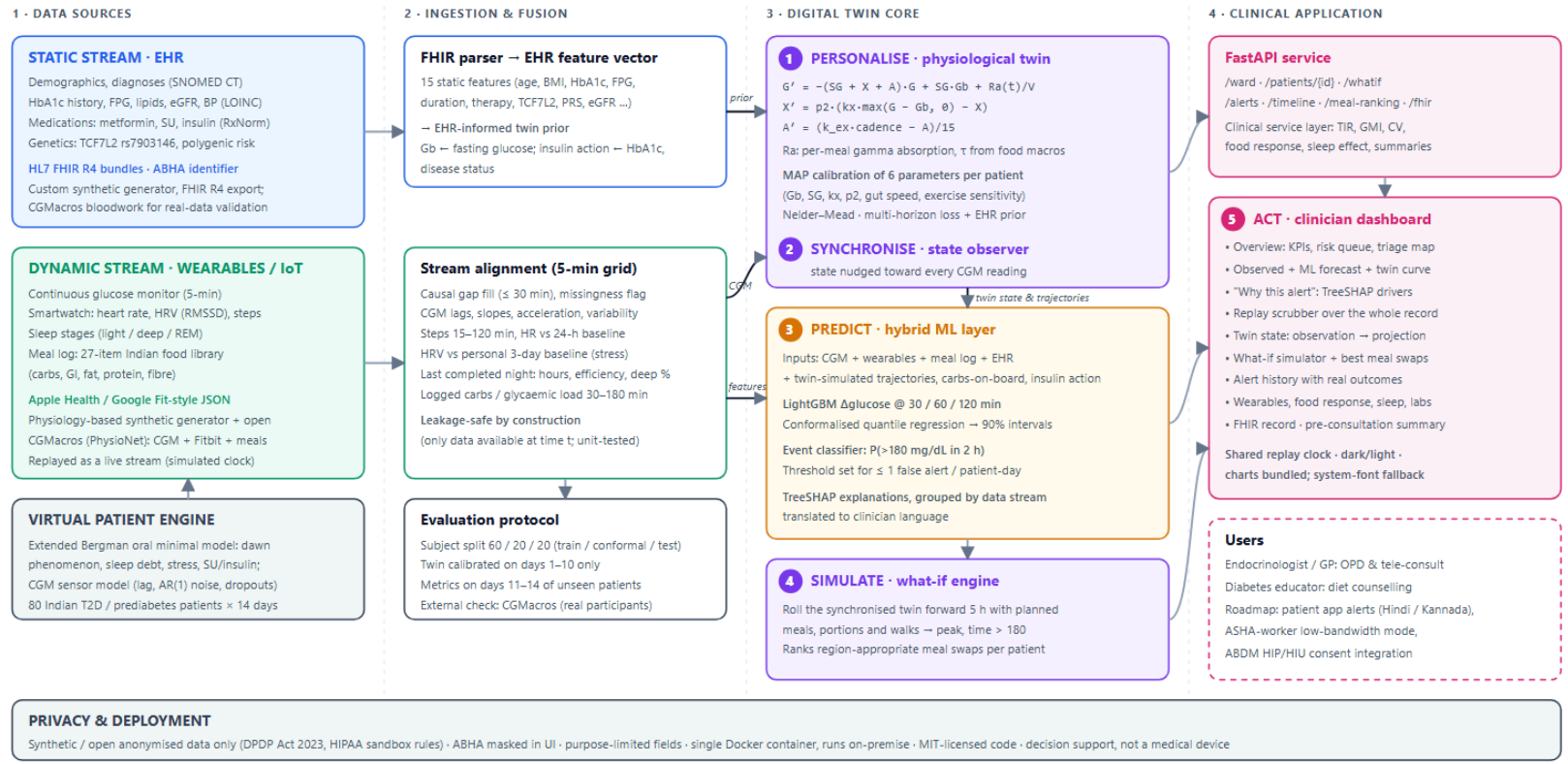
Static: demographics, diagnoses, HbA1c history, labs, medications, TCF7L2 genotype, polygenic risk (HL7 FHIR R4)

Dynamic: CGM, heart rate, HRV, steps, sleep stages, meal log (Apple Health / Google Fit-style)

GlucoTwin India: System Architecture

Hybrid physiological + ML digital twin for Type 2 Diabetes · static EHR stream × dynamic wearable stream → 2-hour adverse-event prediction + clinician what-if simulation

Team Vicimus
Vishwakarma Institute of Technology
Happiest Health Digital Twin Challenge 2026



Data flow: EHR seeds the twin's prior → wearable stream synchronises it every 5 minutes → ML layer corrects what physiology misses (sleep debt, stress, dawn effect) → alerts with reasons and what-if simulations reach the doctor.

Stack: Python 3 · NumPy / SciPy / pandas · LightGBM · scikit-learn · FastAPI + Pydantic · HTML/CSS/JS + Plotly · HL7 FHIR R4 · Docker · pytest

Static stream (EHR) · Dynamic stream (wearables) · Mechanistic twin · Machine learning · Clinical application

github.com/chitrangad-ram-sapate · GlucoTwin India

DATA & THE SANDBOX RULES

Synthetic Indian cohort for development, real people for validation

Synthetic virtual-patient engine

- **80 Indian T2D / prediabetes patients × 14 days**
- Ground-truth physiology: extended Bergman model with dawn phenomenon, sleep-debt insulin resistance, stress, SU / insulin action
- **27-item Indian meal library** with regional (North / South) diets
- CGM sensor model: lag, AR(1) noise, dropouts; imperfect meal logging
- EHR exported as **FHIR R4** (SNOMED, LOINC, RxNorm, ABHA)

Real-world validation: CGMacros

- Open-access PhysioNet dataset, **45 real adults** (15 healthy, 16 prediabetes, 14 T2D)
- Dexcom / Libre CGM + Fitbit heart rate & activity + macro-annotated meals + fasting bloodwork
- Same pipeline, one adapter, **no re-design**

Privacy by design

No real identifiable data · DPDP Act 2023 & HIPAA sandbox rules · ABHA masked in UI · runs on-premise in one container

TWIN CORE

A calibrated physiological model, kept in sync with the patient

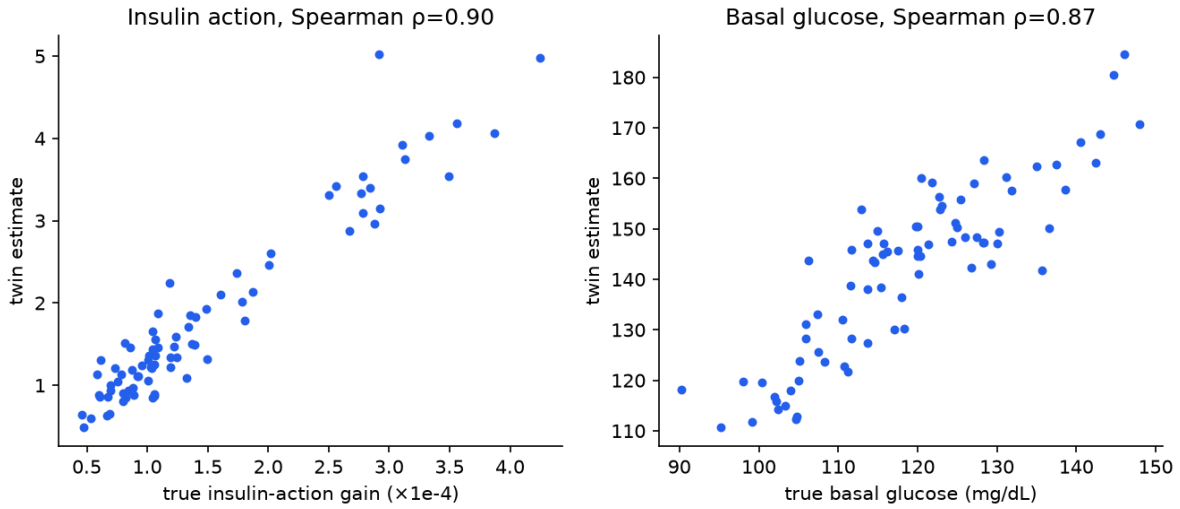
$$G' = -(SG + X + A)\cdot G + SG\cdot Gb + Ra(t)/V$$

$$X' = p2\cdot(kx\cdot\max(G - Gb, 0) - X)$$

$$A' = (k_ex\cdot cadence - A)/15$$

$$Ra = \Sigma \text{ meals } f\cdot carbs\cdot s/\tau^2\cdot e^{-s/\tau}$$

- **EHR prior**: Gb ← fasting glucose, insulin action ← HbA1c and disease status
- **MAP calibration** of Gb, SG, kx, p2, gut speed, exercise sensitivity on days 1–10
- **Observer** corrects state at every CGM reading
- Meal absorption speed derived from **carbs, GI, fat, protein, fibre**



The twin recovers the hidden physiology of each synthetic patient: Spearman ρ = **0.90** (insulin action), **0.87** (basal glucose). Personal calibration cuts the twin's forecast loss by **21%** vs the EHR-only prior.

HYBRID ML LAYER

Machine learning corrects what physiology cannot see

- **70 fused features** per 5-min step: CGM trends, steps, HR, HRV vs personal baseline, last night's sleep, logged carbs and glycaemic load, EHR, **twin-simulated trajectories**, carbs on board, insulin action
- **LightGBM** forecasts Δglucose at 30 / 60 / 120 min
- **Conformalised quantile regression** gives 90% intervals calibrated on held-out patients (measured coverage 89% at 60 min)
- **Event classifier** for spikes (>180) within 2 h. A hypoglycaemia classifier is trained only if the cohort has enough events; this one did not, so no hypo risk is claimed

Alert fatigue is a safety issue

The alert threshold is chosen as the most sensitive operating point with **≤ 1 false alert per patient-day** on held-out patients, not by maximising a statistic. Alerts must persist 10 min and are snoozed for 60 min, like CGM alarms.

Explainable by design

TreeSHAP contributions are grouped by data stream and translated into plain language: "carbs eaten in the last hour", "sleep debt", "HRV below personal baseline".

Leakage-proof

Every feature uses only data available at time t. A unit test perturbs the future and checks that present features do not change.

RESULTS · UNSEEN PATIENTS, UNSEEN DAYS

Every data stream adds value, and the hybrid twin is best

17.5 mg/dL

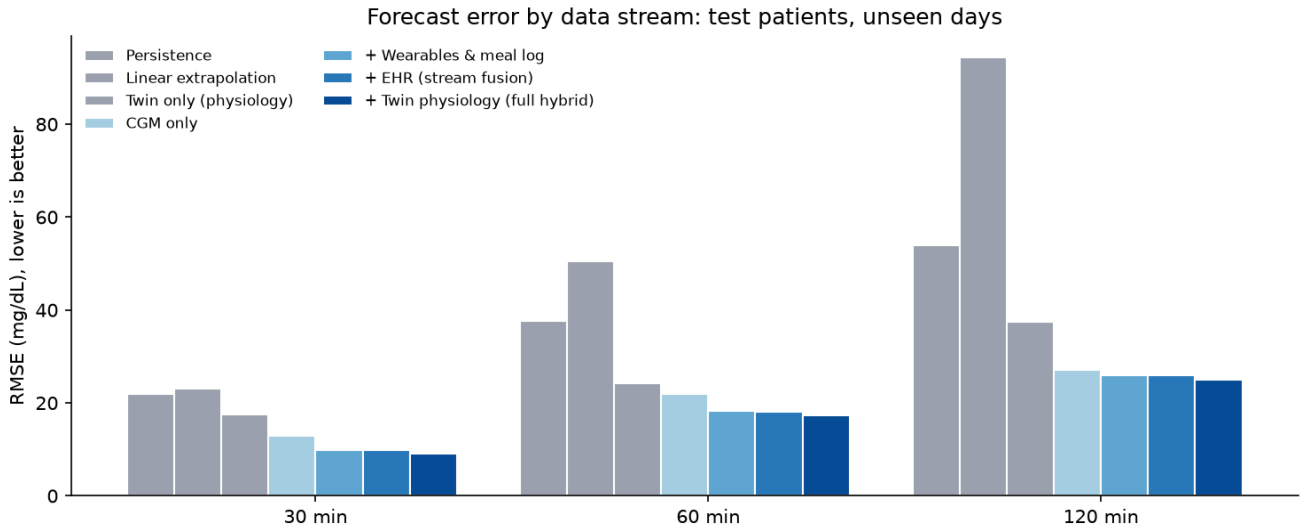
60-min forecast RMSE (persistence: 37.6) · MARD 7.5%

80%

of excursions >180 flagged in advance (129/162), median lead **70 min**, 0.62 false alerts / patient-day

99.7%

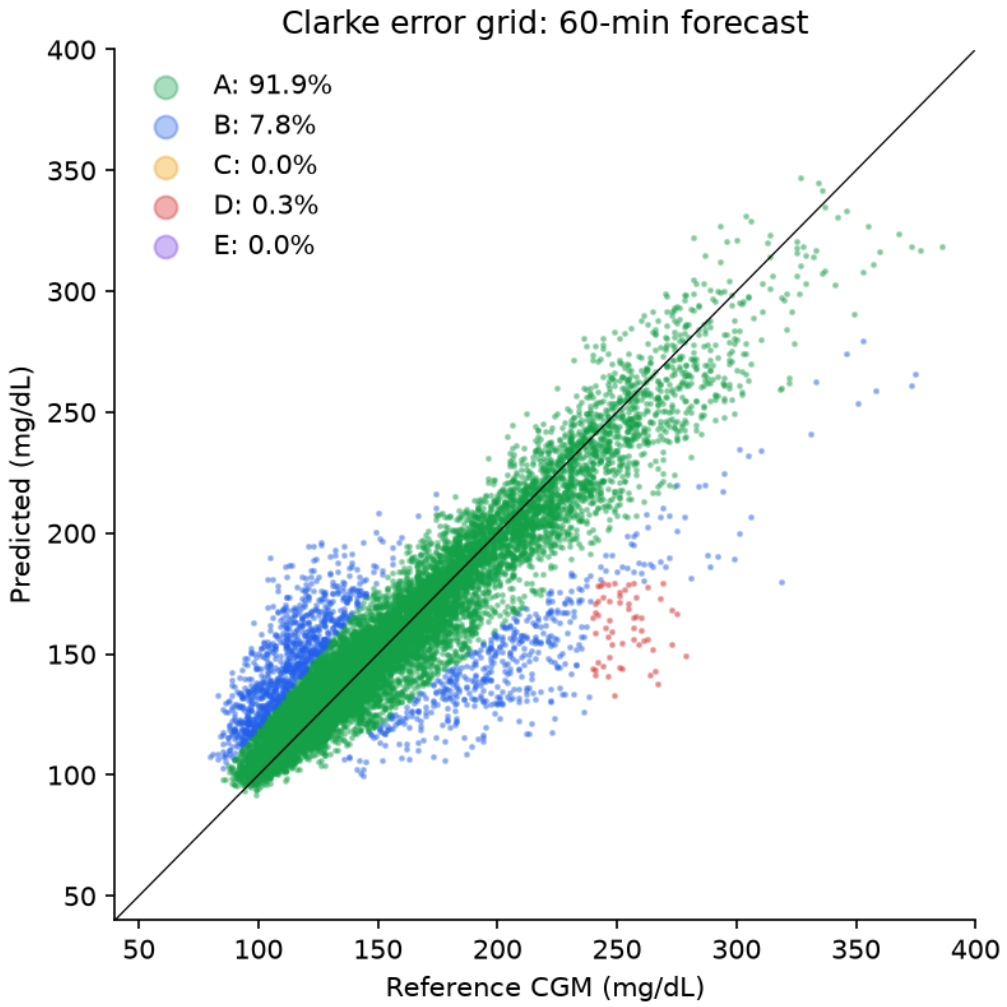
Clarke error grid zones A+B at 60 min · 90% intervals cover 89%

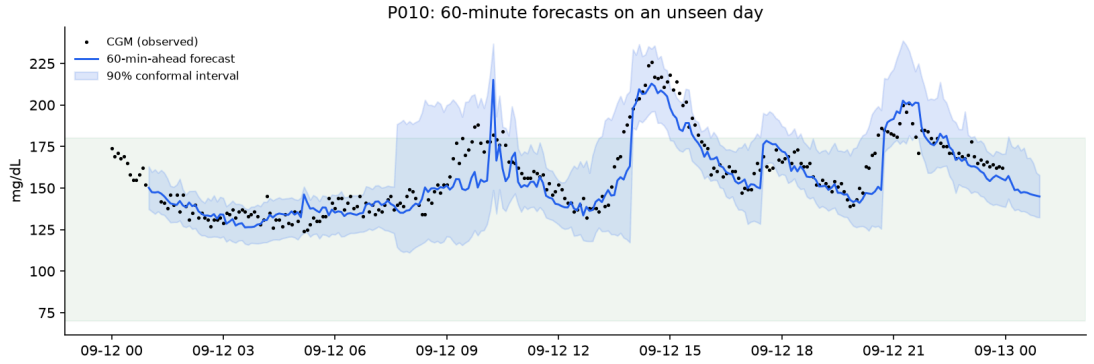


Protocol: subject split 48 / 16 / 16 · twin calibrated on days 1–10 · metrics on days 11–14

CLINICAL ACCURACY

Clinically safe forecasts, with honest uncertainty





Horizon	RMSE	MARD	Clarke A	90% PI coverage
30 min	9.1	4.5%	99.1%	90.0%
60 min	17.5	7.5%	91.9%	88.9%
120 min	25.2	10.7%	84.8%	88.1%

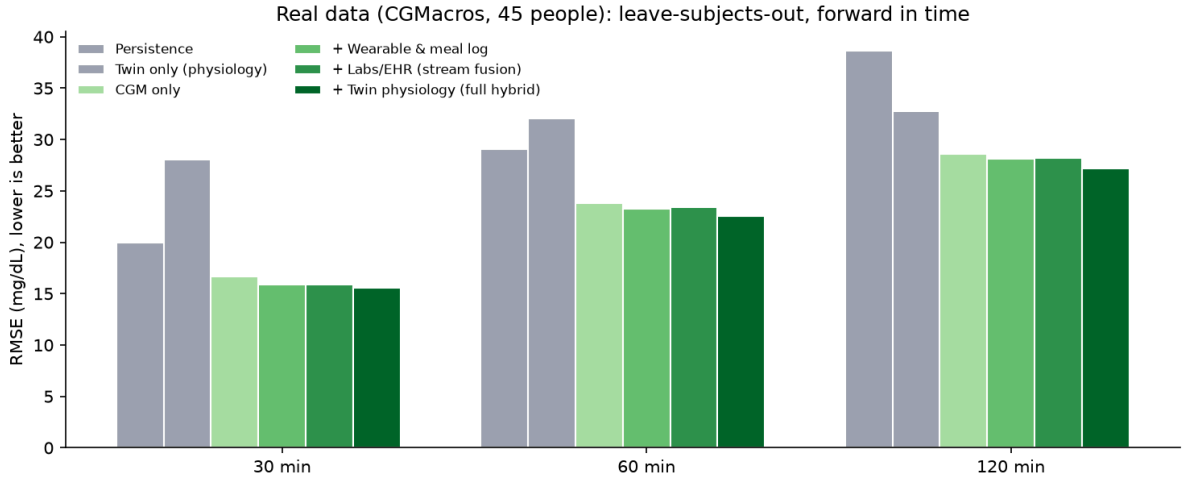
EXTERNAL VALIDATION ON REAL PEOPLE

It works on real CGM data too: 45 adults, CGMacros

Model (leave-subjects-out)	RMSE@30	@60	@120
Persistence	20.0	29.1	38.6
Twin only (physiology)	28.1	32.1	32.7
CGM only	16.6	23.8	28.6
+ Wearable & meal log	15.9	23.2	28.1
+ Labs/EHR (stream fusion)	15.9	23.5	28.2
+ Twin physiology (full hybrid)	**15.5**	**22.5**	**27.2**

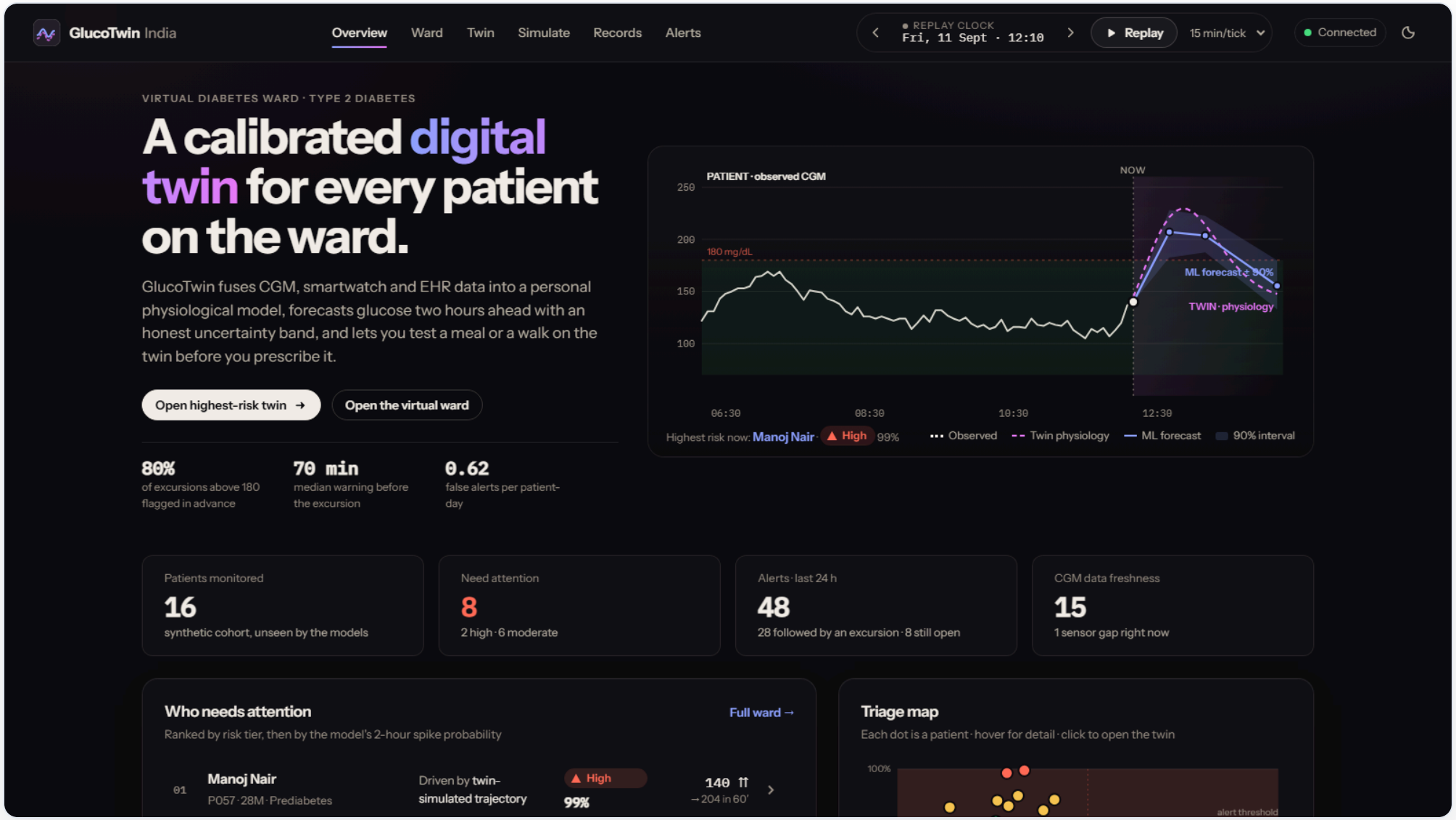
Twin calibrated on the first 60% of each recording; ML never sees the test participants; metrics on the last 40%. Spikes >180 flagged in advance: **207/298**, median lead **41 min**.

No HRV or sleep staging in CGMacros, activity from Fitbit METs, and the cohort is American. The same pipeline ran with a single adapter.



THE DOCTOR'S VIEW

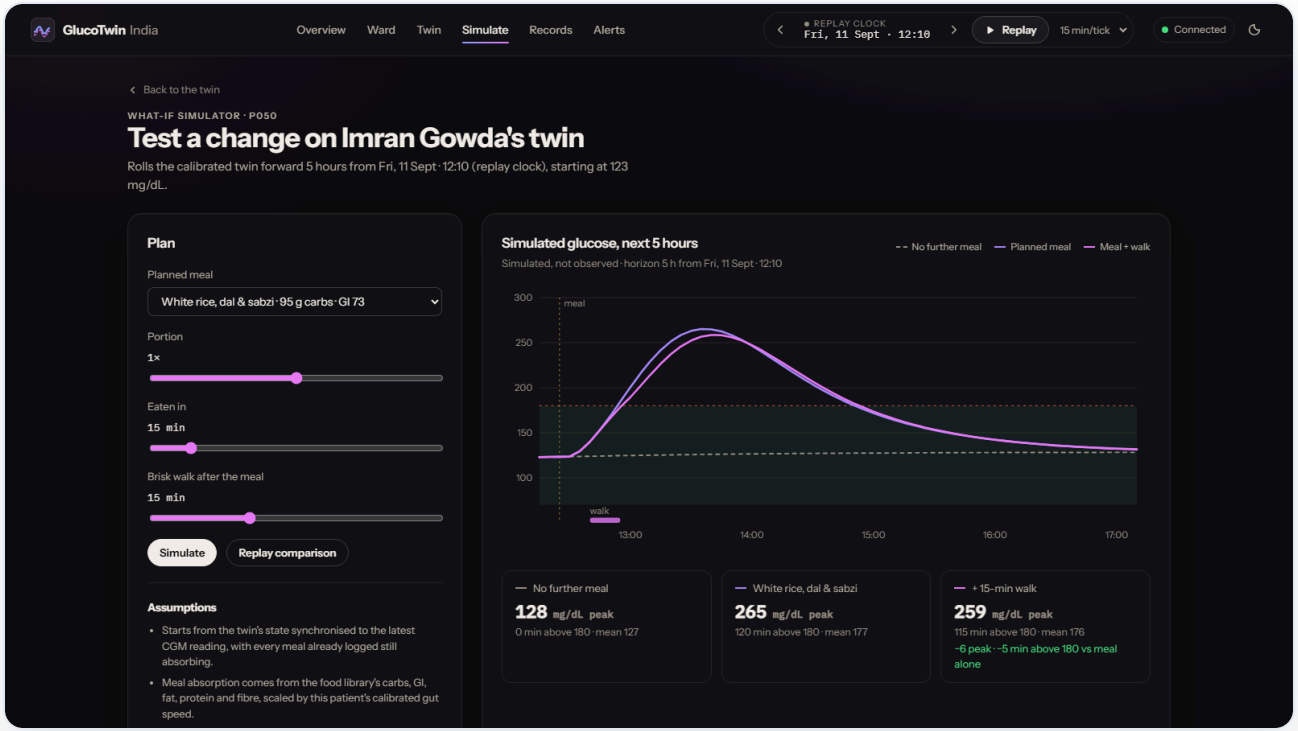
A virtual ward, triaged by 2-hour risk, with the twin of every patient



Risk queue · triage map · twin of the highest-risk patient · shared replay clock (2026-09-11 12:10, synthetic cohort)

WHAT-IF SIMULATOR

Ask the twin before you prescribe



One meal, three futures

No further meal vs. rice-dal-sabzi vs. the same meal + a 15-min walk, simulated from the patient's **current** state.

Best swaps

Ranks region-appropriate alternatives by simulated time above 180 mg/dL **for this patient**.

Personal food response

From Imran Gowda's own logs: Sambar rice with poriyal +147 mg/dL on average (3 meals) vs Roasted chana +10 mg/dL (2).

BUILT FOR INDIA

Designed around Indian patients, food and health infrastructure

🍛 Indian diet

27 dishes (idli, dosa, poha, paratha, rajma chawal, biryani, ragi, jowar, chai-biscuits…) with GI and macros driving absorption.

🧬 South Asian phenotype

Diabetes at lower BMI, higher insulin resistance, TCF7L2 risk alleles, polygenic risk.

🏥 ABDM-ready

HL7 FHIR R4 resources and ABHA identifiers, ready for HIP/HIU consent flows.

🔒 DPDP Act 2023

Synthetic / anonymised data, masking, purpose limitation, on-premise single-container deployment.

Pre-consultation summary

Auto-drafted narrative (TIR, GMI, variability, food response, sleep effect, phenotype) with discussion points, so a 5-minute OPD slot starts informed.

Low-cost path to scale

Works with affordable 14-day CGMs and any smartwatch; next: Hindi / Kannada alerts and an ASHA-worker low-bandwidth mode.

HONEST LIMITATIONS & ROADMAP

What a proof of concept can and cannot claim

Limitations

- Synthetic results are an **upper bound**: we designed that physiology
- CGMacros is small (n = 45), US-based and has no sleep / HRV
- Meal GI not annotated in real data (default used)
- Decision support only; not a medical device

Roadmap

- **Validate** on Indian CGM cohorts with partner diabetologists
- **Twin upgrades**: particle-filter state estimation, medication-titration simulation
- **Patient app**: vernacular alerts, WhatsApp / SMS for ASHA workers
- **ABDM sandbox** integration and a prospective clinic pilot

THANK YOU

From reactive HbA1c reviews to a twin that sees the next two hours.

Team **Vicimus** · Vishwakarma Institute of Technology
Chitrangad Ram Sapate · Soham Joshi

Code (MIT): github.com/chitrangad-ram-sapate/Vicimus_Vishwakarma-Institute-of-Technology
Run: `python -m glucotwin.pipeline && uvicorn glucotwin.api:app`